

03 • L1 — Guidelines & Evidence

Assignment: “Guidelines and Evidence — identify relevant clinical guidelines and evidence and briefly describe the rationale for selection.”

3.1 L1 architecture: three roles, four sources

We deliberately distinguish between the **normative basis** (what constitutes an infection, when action is required), the **assessment instrument** (how the wound appearance is described) and the **evidence on instrument choice** (how well these instruments actually perform).

Role	Source	What it provides	Type of evidence
Normative — definition	CDC/NHSN Surgical Site Infection Event	Definition of superficial / deep incisional SSI, time windows, clinical criteria	International surveillance definition, consensus-based, updated annually
Normative — action	NICE NG125 <i>Surgical site infections: prevention and treatment</i> (11 April 2019, last updated 19 August 2020)	Postoperative wound care, patient education on signs of infection, treatment in case of suspicion	Evidence-based guideline with systematic appraisal of the evidence
Instrument	Southampton Wound Scoring System (Bailey et al.)	Visual grading of the wound appearance in stages 0–V with sub-grades	Validated, published assessment instrument
Evidence	Campwala, Unsell & Gupta (2019)	Comparison of the predictive value of CDC, ASEPSIS and Southampton	Retrospective cohort study, n = 22

3.2 Rationale for the selection

CDC/NHSN SSI criteria — why

The CDC/NHSN definition is the international reference standard for SSI surveillance. It provides three things that no other source specifies with comparable precision:

1. **The time window:** 30 days after the procedure (90 days for implant-related procedures) — this justifies the duration of our follow-up period.
2. **The cardinal clinical signs** of superficial incisional SSI: “*new or worsening localized pain or tenderness; localized swelling; erythema; or heat*” as well as “*purulent drainage from the superficial incision*”. These five signs form the backbone of our data dictionary.
3. **The distinction** between superficial and deep incisional SSI — relevant for us in differentiating ORANGE (superficial suspicion) from RED (dehiscence, deep involvement).

Deliberate limitation: The CDC definition partly presupposes a medical examination, culture results or active reopening of the wound. These criteria cannot be captured during self-monitoring at home. We therefore adopt **only the clinical and visual criteria** and derive a *suspicion* from them — not a diagnosis. See Challenge C-05.

NICE NG125 — why

NG125 is a methodologically rigorous, evidence-based guideline with a systematic appraisal of the evidence and is current (2019, updated 2020). Particularly relevant for us:

- **Recommendation 1.1.3** explicitly requires that patients and their relatives receive guidance on “*how to recognise a surgical site infection and who to contact if they are concerned*”. **This is the guideline-based legitimization of our entire use case** — WundCheck operationalises a recommendation that NICE already makes, but whose implementation in practice usually remains a leaflet.
- **Recommendations 1.4.1–1.4.3** on postoperative wound care — aseptic non-touch technique for dressing changes (1.4.1), sterile saline for cleansing up to 48 h (1.4.2), showering permitted from 48 h (1.4.3) — feed into our information rules.
- **Recommendation 1.4.9** on treatment in suspected cellulitis justifies why a spreading erythema is treated as a warning criterion in its own right.

Southampton Wound Scoring System — why

The decisive reason: **Southampton is a visual grading system**. Our central interaction is an image selection. Without an established instrument, our staging would be a purely self-devised construct without a source — with Southampton it is anchored to a published system that is widely used in SSI research.

In addition, Southampton provides a meaningful ordinal scale with a clear severity gradient (0 → V) that maps directly onto a traffic light, as well as sub-grades for extent and duration that structure our follow-up questions.

Campwala et al. 2019 — why, and with which limitation

Full citation: Campwala I, Unsell K, Gupta S. *A Comparative Analysis of Surgical Wound Infection Methods: Predictive Values of the CDC, ASEPSIS, and Southampton Scoring Systems in Evaluating Breast Reconstruction Surgical Site Infections*. Plastic Surgery (Oakville). 2019. PMID 31106164.

Characteristic	Value
Study design	Retrospective, single-centre
Sample	n = 22 cases with postoperative wound complications
Population	Implant-based breast reconstruction, 01/2013–06/2016
Endpoint	Failure of the implant-based reconstruction

Results:

- An ASEPSIS score > 30 correlated with a > 50 % failure rate (p = .022)
- Southampton class B: 60 % failure · class C: 23 % · class D: 0 %
- **CDC scoring had no predictive value** for success vs. failure

Authors' conclusion: ASEPSIS demonstrated substantial predictive value; the CDC criteria and Southampton scoring showed limited clinical utility.

3.3 Appraisal of the evidence

Day 2 of the module explicitly requires an appraisal of the L1 sources and their level of evidence. This section is

our response to that requirement.

Source	Level of evidence	Strengths	Limitations for our use case
CDC/NHSN	Consensus-based surveillance definition, no GRADE rating	Internationally established, precisely operationalised, updated regularly	Designed for surveillance , not for point-of-care triage. Presupposes professional judgement and, in part, laboratory diagnostics
NICE NG125	Evidence-based guideline, systematic appraisal of the evidence	Methodologically rigorous, current, contains an explicit patient-education recommendation	British care context; recommendations are addressed to health professionals, not to patients
Southampton	Validated instrument, moderate evidence base	Visual, ordinal, suitable for image selection	Developed for clinical assessment , not for lay self-assessment
Campwala et al.	Retrospective, n = 22, single-centre — low level of evidence	Direct comparison of the three instruments	Very small sample; the population (implant-based breast reconstruction) differs substantially from our scope; the endpoint is implant failure, not detection of infection

The critical appraisal

Campwala et al. (2019) show that the three common instruments perform differently — with Southampton showing only limited predictive value. The study is retrospective and single-centre, with n = 22 implant-based breast reconstructions. Its transferability to our broader population (all elective wounds with primary closure, all body regions) is correspondingly limited.

We therefore deliberately use Southampton not as a prognostic instrument, but as a visual communication and triage grid. For our objective — timely contact with a health professional rather than prediction of the healing course — comprehensibility for lay users matters more than predictive accuracy. An instrument that predicts the healing course well but cannot be applied by patients would be of no value to us.

We **deliberately rejected ASEPSIS**, even though it performed best in this study: ASEPSIS requires the recording of antibiotic administration, drainage, wound debridement, hospital stay and culture results over several days — none of which patients can capture at home. This is a decision against the statistically superior option in favour of the **applicable** option, and we state it openly.

3.4 Southampton Mapping

Assignment of the Southampton grades to our wound images and to the traffic-light classification. This table is the bridge between L1 and L2.

Grade	Southampton definition (original)	Our image	Data elements	Traffic light
0	Normal healing	wunde_1_reizlos	wound_state = 1	 GREEN
I	Normal healing with mild bruising and/or erythema <i>Ia: some bruising · Ib: considerable bruising · Ic: mild erythema</i>	wunde_2_leicht_geroetet	wound_state = 2	 GREEN (day 1–3)  YELLOW (from day 4)

Grade	Southampton definition (original)	Our image	Data elements	Traffic light
II	Erythema plus other signs of inflammation <i>Ila: at one point · IIb: around sutures · IIc: along wound · IId: around wound</i>	wunde_3_stark_geroetet	wound_state = 3 , redness_extent , swelling , warmth	 YELLOW /  ORANGE
III	Clear / serous / bloody discharge <i>IIIa: ≤ 2 cm · IIIb: > 2 cm along wound · IIIc: large volume · IIId: prolonged > 3 days</i>	sekret_1_klar , sekret_2_blutig	secretion = 1 2 , discharge_extent , discharge_duration_days	 ORANGE
IV	Pus <i>IVa: ≤ 2 cm · IVb: > 2 cm along wound</i>	sekret_3_eitrig , wunde_4_offen_eiter	secretion = 3	 ORANGE
V	Deep / severe wound infection ± tissue breakdown	wunde_4_offen_eiter	wound_state = 4 , wound_open = 2	 RED

Grades **IV** and **V** count as **major complications** in the Southampton system.

 **Correction after the test run (bug B-06):** grade IV was originally mapped to RED here, while the decision table classified isolated purulent discharge as ORANGE (R4). The rules are authoritative: purulent discharge requires contact **today**, not the emergency service tonight. RED remains reserved for systemic signs (R1), lymphangitis (R2) and dehiscence or the “open / pus” wound image (R3). Rationale: challenge C-09.

The derived data element `southampton_grade` (DE-030) computes the grade from `wound_state` , `redness_extent` , `swelling` , `secretion` and `wound_open` — making it our most clinically informative derived element.

3.5 Adaptation decisions

Three decisions were necessary to make Southampton usable for lay self-assessment. All three are documented in full in 07 L1→L2 challenges:

1 • Sub-grades without centimetre measurements (→ C-02) Southampton distinguishes “≤ 2 cm” from “> 2 cm along the wound”. Patients do not measure at home. Instead we ask: “*At one small spot or along the whole wound?*” (“An einer kleinen Stelle oder entlang der ganzen Wunde?”) — semantically close, but without measurement. The price: lower precision and possible systematic underestimation.

2 • Grade I is time-dependent (→ C-03) Mild erythema on day 2 is a physiological inflammatory response; on day 8 it is a warning sign. Southampton does not account for this temporal dimension. We link Grade I to `days_post_op` with day 3 as the cut-off. The price: the threshold is set by us, not derived from the source.

3 • Professional instrument → lay instrument (→ C-01) The most fundamental decision: Southampton was developed for assessment by trained personnel. We translate it into an image selection for lay users. This is the core of our project — and at the same time our greatest untested risk, because the reliability of lay image matching has not been validated. See the residual-risk statement in 10 QA testing.

3.6 Sources

- **CDC/NHSN.** *Surgical Site Infection Event (SSI). Patient Safety Component Manual.* <https://www.cdc.gov/nhsn/pdfs/pscmanual/9pscscsscurrent.pdf>
 - **NICE.** *Surgical site infections: prevention and treatment.* NICE guideline NG125, 11 April 2019, last updated 19 August 2020. <https://www.nice.org.uk/guidance/ng125>
 - **Bailey IS et al.** Southampton Wound Scoring System. — Table reproduced in: *Use of Southampton Scoring for Wound Healing in Post-surgical Patients.* <https://cdn.fortunejournals.com/articles/use-of-southampton-scoring-for-wound-healing-in-postsurgical-patients.pdf>
 - **Campwala I, Unsell K, Gupta S.** *A Comparative Analysis of Surgical Wound Infection Methods: Predictive Values of the CDC, ASEPSIS, and Southampton Scoring Systems in Evaluating Breast Reconstruction Surgical Site Infections.* Plastic Surgery. 2019. PMID 31106164. <https://journals.sagepub.com/doi/10.1177/2292550319826095>
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Related documents: 05 Data Dictionary • 06 Decision Logic • 07 L1→L2 Challenges